

Chapter 1

Plane

Coordinate

Geometry

Chapter 1 Plane Coordinate Geometry

The Coordinate Plane

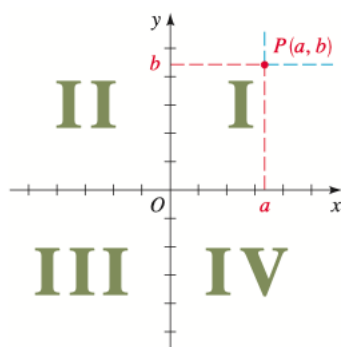


FIGURE 1

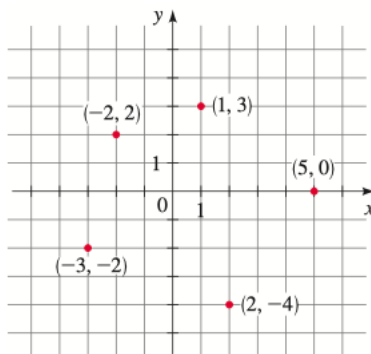


FIGURE 2

EXAMPLE 1

Graphing Regions in the Coordinate Plane

Describe and sketch the regions given by each set.

- (a) $\{(x, y) \mid x \geq 0\}$
- (b) $\{(x, y) \mid y = 1\}$
- (c) $\{(x, y) \mid |y| < 1\}$

SOLUTION

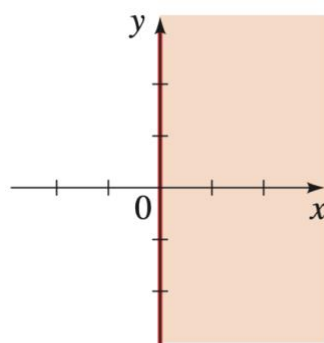
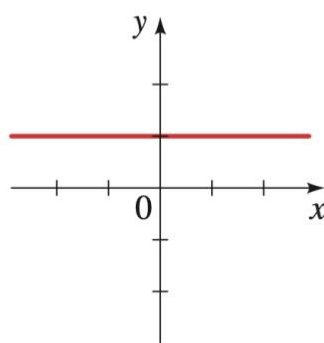
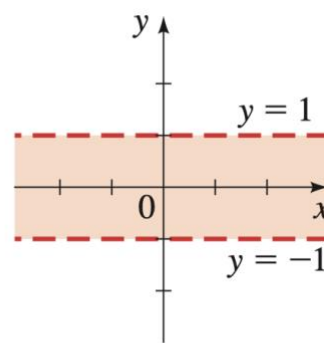
(a) $x \geq 0$ (b) $y = 1$ (c) $|y| < 1$

FIGURE 3

The Distance and Midpoint Formulas

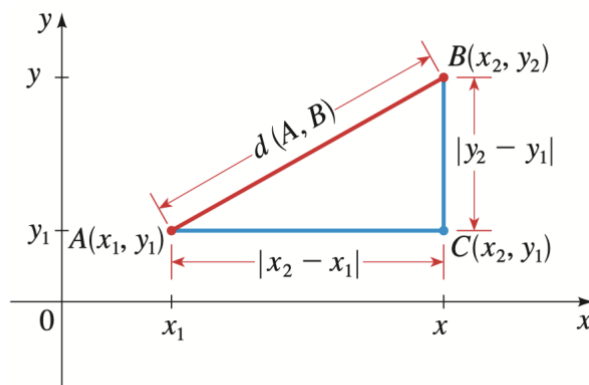


FIGURE 4

DISTANCE FORMULA

The distance between the points $A(x_1, y_1)$ and $B(x_2, y_2)$ in the plane is

$$d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

EXAMPLE 2 Applying the Distance Formula

Which of the points $P(1, -2)$ or $Q(8, 9)$ is closer to the point $A(5, 3)$?

Justify your answer.

SOLUTION

$$\begin{aligned} d(P, A) &= \sqrt{(5-1)^2 + (3-(-2))^2} \\ &= \sqrt{4^2 + 5^2} = \sqrt{41} \end{aligned}$$

$$\begin{aligned} d(Q, A) &= \sqrt{(5-8)^2 + (3-9)^2} \\ &= \sqrt{(-3)^2 + (-6)^2} \\ &= \sqrt{45} \end{aligned}$$

$$d(P, A) < d(Q, A)$$

P is closer to A

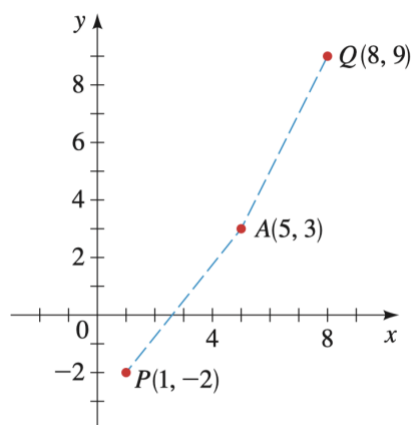


FIGURE 5

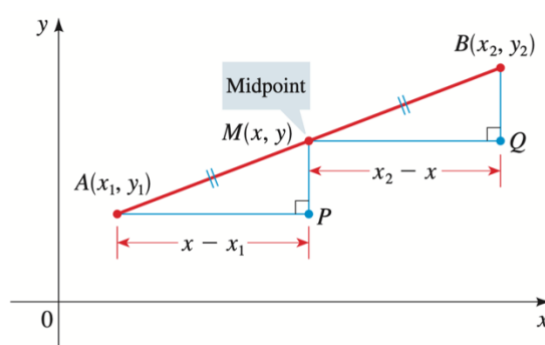


FIGURE 6

MIDPOINT FORMULA

The midpoint of the line segment from $A(x_1, y_1)$ to $B(x_2, y_2)$ is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

EXAMPLE 3 Applying the Midpoint Formula

Show that the quadrilateral with vertices $P(1, 2)$, $Q(4, 4)$, $R(5, 9)$, and $S(2, 7)$ is a parallelogram by proving that its two diagonals bisect each other.

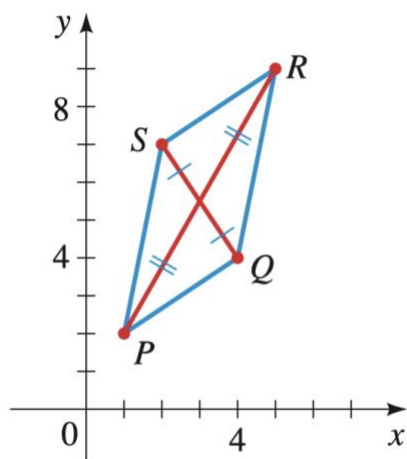
SOLUTION If the two diagonals have the same midpoint, then they must bisect each other. The midpoint of the diagonal PR is

$$\left(\frac{1 + 5}{2}, \frac{2 + 9}{2} \right) = \left(3, \frac{11}{2} \right)$$

and the midpoint of the diagonal QS is

$$\left(\frac{4 + 2}{2}, \frac{4 + 7}{2} \right) = \left(3, \frac{11}{2} \right)$$

so each diagonal bisects the other, as shown in Figure 7. (A theorem from elementary geometry states that the quadrilateral is therefore a parallelogram.)

**FIGURE 7**

Fundamental Principle of Analytic Geometry

A point (x, y) lies on the graph of an equation if and only if its coordinates satisfy the equation.

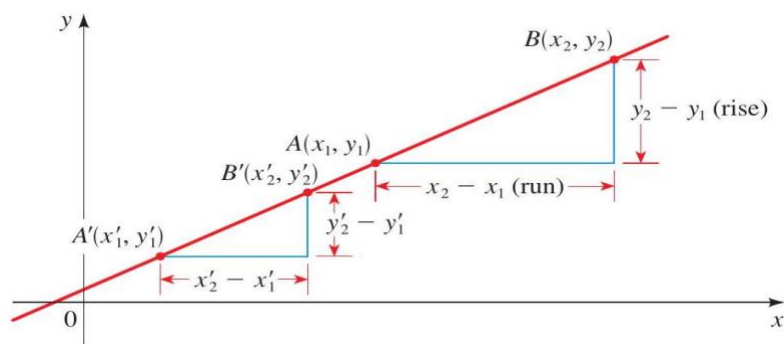
LINES

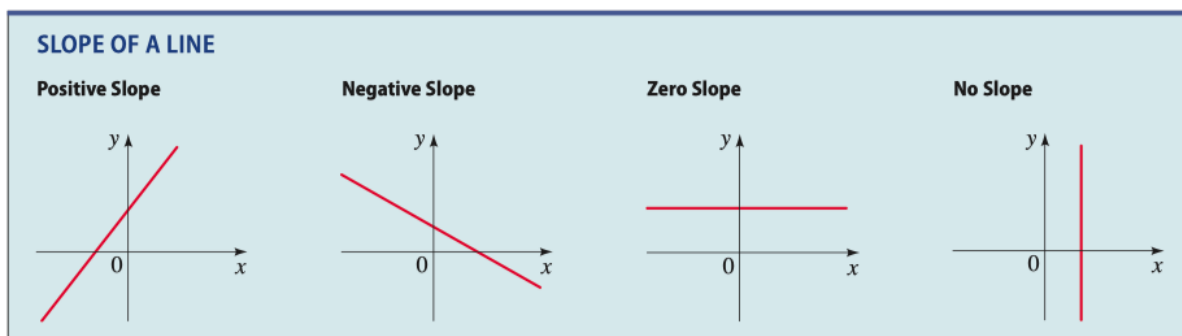
The slope m of a nonvertical line that passes through the points $A(x_1, y_1)$ and $B(x_2, y_2)$ is

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

The slope of a vertical line is not defined.

The slope is independent of which two points are chosen on the line. We can see that this is true from the similar triangles.



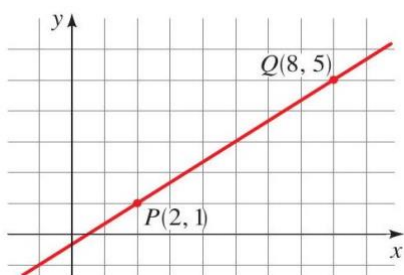


EXAMPLE 4 Finding the Slope of a Line Through Two Points

Find the slope of the line that passes through the points $P(2,1)$ and $Q(8,5)$.

SOLUTION Since any two different points determine a line, only one line passes through these two points. From the definition the slope is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5 - 1}{8 - 2} = \frac{4}{6} = \frac{2}{3}$$



Point-Slope Form of the Equation of a Line

$$\frac{y - y_1}{x - x_1} = m$$

An equation of the line that passes through the point (x_1, y_1) and has slope m is

$$y - y_1 = m(x - x_1)$$

EXAMPLE 5 Finding an Equation of a Line with Given Point and Slope

(a) Find an equation of the line through $(1, -3)$ with slope $-\frac{1}{2}$.

(b) Sketch the line.

SOLUTION

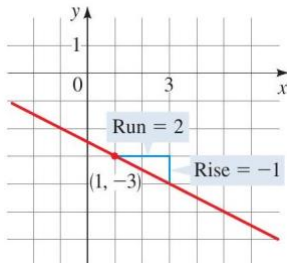
(a) Using the point-slope form with $m = -\frac{1}{2}$, $x_1 = 1$, and $y_1 = -3$, we obtain an equation of the line as

$$y - y_1 = m(x - x_1)$$

$$y - (-3) = -\frac{1}{2}(x - 1)$$

$$\begin{array}{ll}
 y + 3 = -\frac{1}{2}(x - 1) & \text{Slope } m = -\frac{1}{2}, \text{ point } (1, -3) \\
 2y + 6 = -x + 1 & \text{Multiply by 2} \\
 x + 2y + 5 = 0 & \text{Rearrange}
 \end{array}$$

(b) The fact that the slope is $-\frac{1}{2}$ tells us that when we move to the right 2 units, the line drops 1 unit. This enables us to sketch the line in Figure 6.



EXAMPLE 5 Finding an Equation of a Line Through Two Given Points

Find an equation of the line through the points $(-1, 2)$ and $(3, -4)$.

SOLUTION

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-4 - 2}{3 - (-1)} = \frac{-6}{4} = -\frac{3}{2}$$

$$y - y_1 = m(x - x_1)$$

$$y - 2 = -\frac{3}{2}(x - (-1))$$

$$2y - 4 = -3x - 3$$

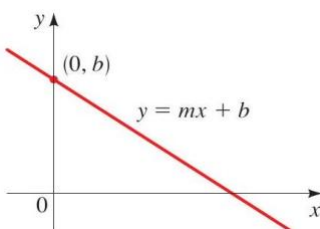
$$3x + 2y - 1 = 0$$

$$ax + by + c = 0$$

Slope-Intercept Form of the Equation of a Line

$$y - b = m(x - 0)$$

This simplifies to $y = mx + b$, which is called the slope-intercept form of the equation of a line.



An equation of the line that has slope m and y -intercept b is

$$y = mx + b$$

EXAMPLE 6 Lines in Slope-Intercept Form

(a) Find an equation of the line with slope 3 and y -intercept -2 .

(b) Find the slope and y -intercept of the line $3y - 2x = 1$.

SOLUTION

(a) Since $m = 3$ and $b = -2$, from the slope-intercept form of the equation of a line we get

$$y = mx + b$$

$$y = 3x - 2$$

(b) We first write the equation in the form $y = mx + b$.

$$3y - 2x = 1$$

$$3y = 2x + 1$$

$$y = \frac{2}{3}x + \frac{1}{3}$$

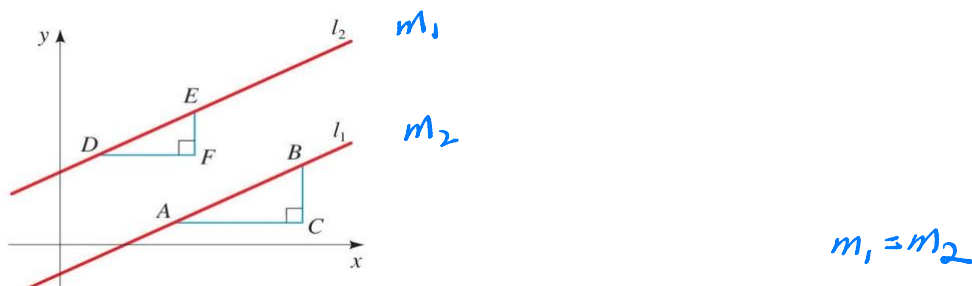
$$y = mx + b$$

$$m = \frac{2}{3} = \text{slope}$$

$$b = \frac{1}{3} = \text{y-intercept}$$

Parallel and Perpendicular Lines

PARALLEL LINES



Two nonvertical lines are parallel if and only if they have the same slope.

EXAMPLE 7 Finding an Equation of a Line Parallel to a Given Line

Find an equation of the line through the point (5,2) that is parallel to the line $4x + 6y + 5 = 0$.

SOLUTION First we write the equation of the given line in slope-intercept form.

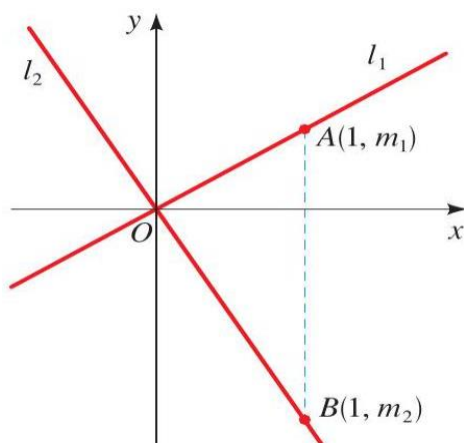
$4x + 6y + 5 = 0$ $6y = -4x - 5$ $y = -\frac{4}{6}x - \frac{5}{6}$ $y = -\frac{2}{3}x - \frac{5}{6}$ $y = mx + b$ $m = -\frac{2}{3}$	$y - y_1 = m(x - x_1)$ $y - 2 = -\frac{2}{3}(x - 5)$ $3y - 6 = -2x + 10$ $2x + 3y - 16 = 0$ $ax + by + c = 0$
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PERPENDICULAR LINES

Two lines with slopes m_1 and m_2 are perpendicular if and only if $m_1 m_2 = -1$, that is, their slopes are negative reciprocals:

$$m_2 = -\frac{1}{m_1}$$

Also, a horizontal line (slope 0) is perpendicular to a vertical line (no slope).



EXAMPLE 8 Finding an Equation of a Line Perpendicular to a Given Line

Find an equation of the line that is perpendicular to the line $4x + 6y + 5 = 0$ and passes through the origin.

$$6y = -4x - 5$$

$$y = -\frac{4}{6}x - \frac{5}{6}$$

$$y = -\frac{2}{3}x - \frac{5}{6}$$

$$m_1 = -\frac{2}{3}$$

$$m_2 = -\frac{1}{m_1}$$

$$= -\frac{1}{(-\frac{2}{3})}$$

$$= \frac{3}{2}$$

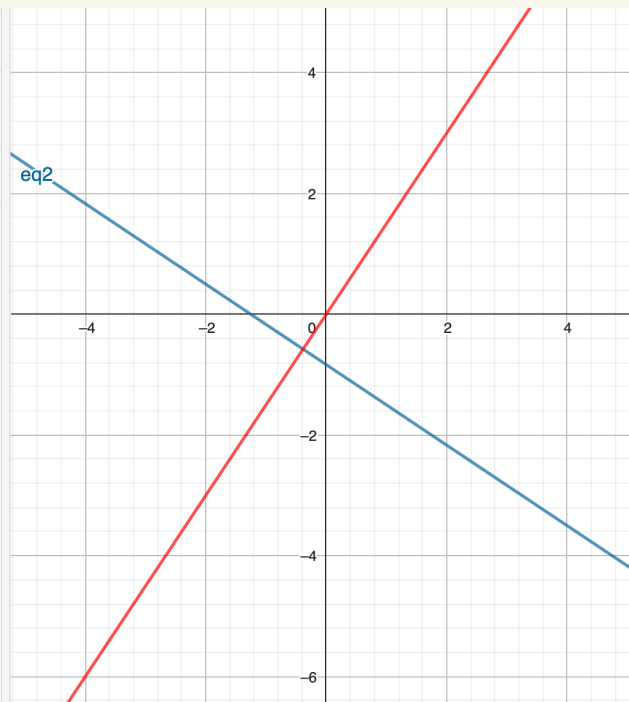
$$y - 0 = m_2(x - 0)$$

$$y = \frac{3}{2}x$$

$\therefore$ The equation that is perpendicular to the line $4x + 6y + 5 = 0$ is

$$y = \frac{3}{2}x$$

●	eq2 : $4x + 6y + 5 = 0$	⋮
●	h : $y = \frac{3}{2}x$	⋮
+	Input...	



$$y = \frac{3}{2}x$$

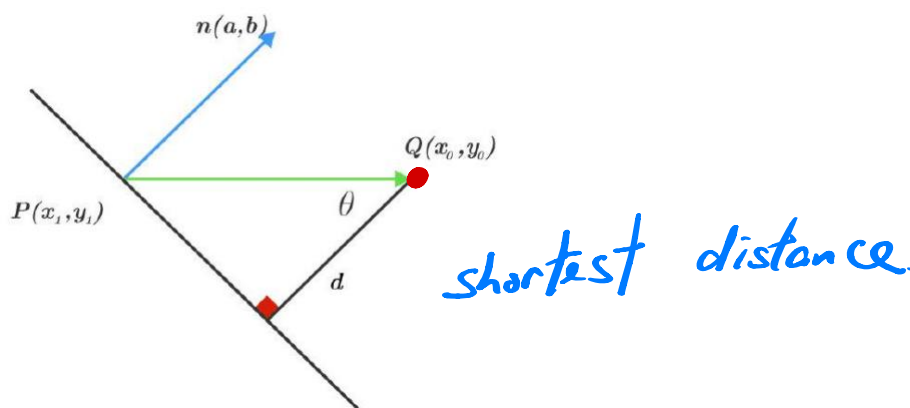
$$4x + 6y + 5 = 0$$

EXAMPLE 8 Finding an Equation of a Line Perpendicular to a Given Line

Find an equation of the line that is perpendicular to the line $4x + 6y + 5 = 0$ and passes through the origin.

SOLUTION In Example 7 we found that the slope of the line $4x + 6y + 5 = 0$ is $-\frac{2}{3}$. Thus the slope of a perpendicular line is the negative reciprocal, that is, $\frac{3}{2}$. Since the required line passes through $(0,0)$, the point-slope form gives

$$\begin{aligned} y - 0 &= \frac{3}{2}(x - 0) & \text{Slope } m &= \frac{3}{2}, \text{ point } (0,0) \\ y &= \frac{3}{2}x & \text{Simplify} \end{aligned}$$

Distance between Point and a Line**THEOREM**

The distance d from a point $Q(x_0, y_0)$ to the line $ax + by + c = 0$ is

$$d = \frac{|a(x_0) + b(y_0) + c|}{\sqrt{a^2 + b^2}}$$

EXAMPLE 9

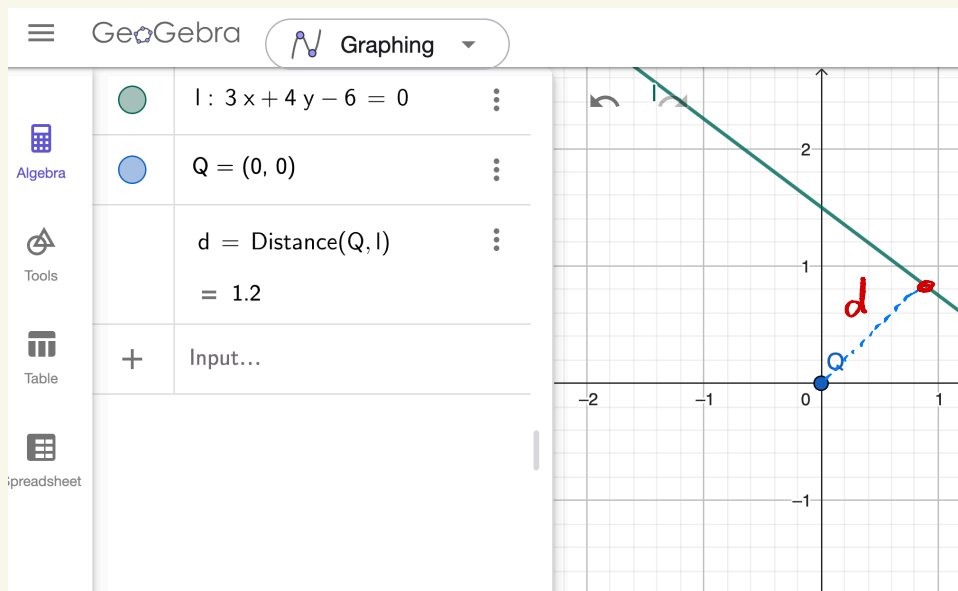
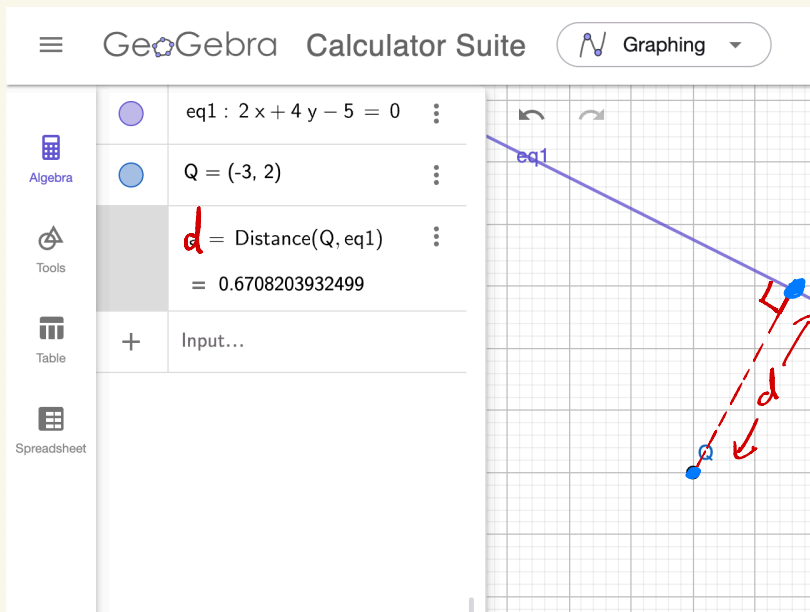
$$ax + by + c = 0 \quad (x_0, y_0)$$

Find the distance between the line l , $2x + 4y - 5 = 0$ and the point $Q = (-3, 2)$,

From the distance formula we have:

$$d = \frac{|2(-3) + 4(2) - 5|}{\sqrt{2^2 + 4^2}} = \frac{3}{2\sqrt{5}} \approx 0.67$$

$$a = 2 \quad b = 4 \quad c = -5 \quad x_0 = -3 \quad y_0 = 2$$



EXAMPLE 10

$$ax + by + c = 0$$

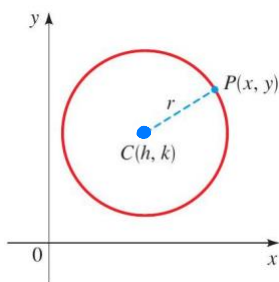
$$(x_0, y_0) \quad \begin{matrix} x_0 = 0 \\ y_0 = 0 \end{matrix}$$

Find the distance between the line l , $3x + 4y - 6 = 0$ and the point $(0,0)$.

$$d = \frac{|a(x_0) + b(y_0) + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|-6|}{\sqrt{3^2 + 4^2}} = \frac{6}{\sqrt{25}} = \frac{6}{5} = 1.2$$

Circles



$$r = \sqrt{(x-h)^2 + (y-k)^2}$$

$$r^2 = (x-h)^2 + (y-k)^2$$

EQUATION OF A CIRCLE

An equation of the circle with center (h, k) and radius r is

$$(x - h)^2 + (y - k)^2 = r^2$$

This is called the standard form for the equation of the circle. If the center of the circle is the origin $(0,0)$, then the equation is

 h, k

$$x^2 + y^2 = r^2$$

EXAMPLE 11 Graphing a Circle

Graph each equation.

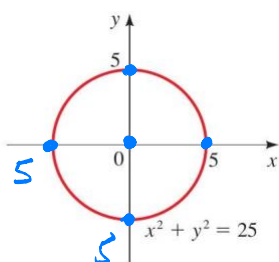
(a) $x^2 + y^2 = 25$

(b) $(x - 2)^2 + (y + 1)^2 = 25$

$$(x - 2)^2 + (y - (-1))^2 = 5^2$$

SOLUTION

(a) Rewriting the equation as $x^2 + y^2 = 5^2$, we see that this is an equation of the circle of radius 5 centered at the origin.



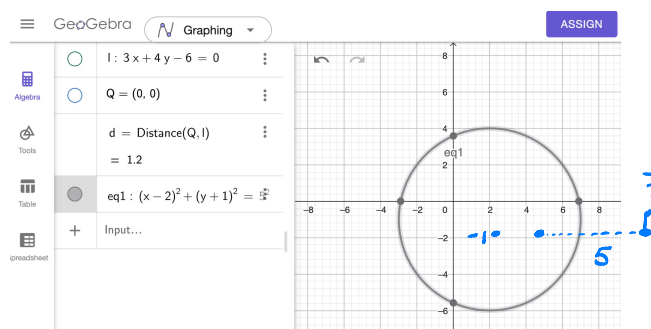
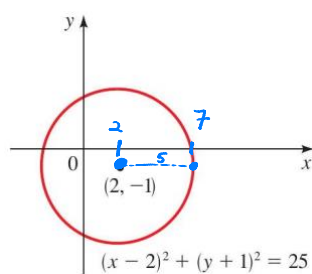
$$C(0,0)$$

$$r^2 = 25$$

$$r = \sqrt{25} = 5$$

$$(x-2)^2 + (y-(-1))^2 = 5^2 \quad 11$$

(b) Rewriting the equation as $(x-2)^2 + (y+1)^2 = 5^2$, we see that this is an equation of the circle of radius 5 centered at $(2, -1)$.



EXAMPLE 12 Finding an Equation of a Circle

(a) Find an equation of the circle with radius 3 and center $(2, -5)$.

(b) Find an equation of the circle that has the points $P(1, 8)$ and $Q(5, -6)$ as the endpoints of a diameter.

$$(x-h)^2 + (y-k)^2 = r^2$$

SOLUTION

(a) Using the equation of a circle with $r = 3$, $h = 2$, and $k = -5$, we obtain

$$(x-2)^2 + (y+5)^2 = 9$$

$$(x-2)^2 + (y-(-5))^2 = 3^2$$

(b) We first observe that the center is the midpoint of the diameter PQ , so by the Midpoint Formula the center is

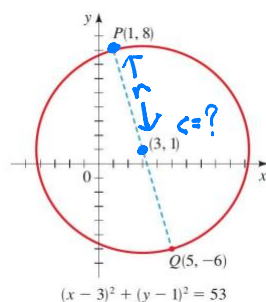
$$C = \left(\frac{1+5}{2}, \frac{8-6}{2} \right) = (3, 1) \quad P(1, 8)$$

The radius r is the distance from P to the center, so by the Distance Formula

$$r^2 = (3-1)^2 + (1-8)^2 = 2^2 + (-7)^2 = 53$$

Therefore the equation of the circle is

$$(x-3)^2 + (y-1)^2 = 53$$



$$(x-h)^2 + (y-k)^2 = r^2$$

standard form

$$C(3, 1) \quad r = \sqrt{53}$$

Let's expand the equation of the circle in the preceding example.

simple ↓

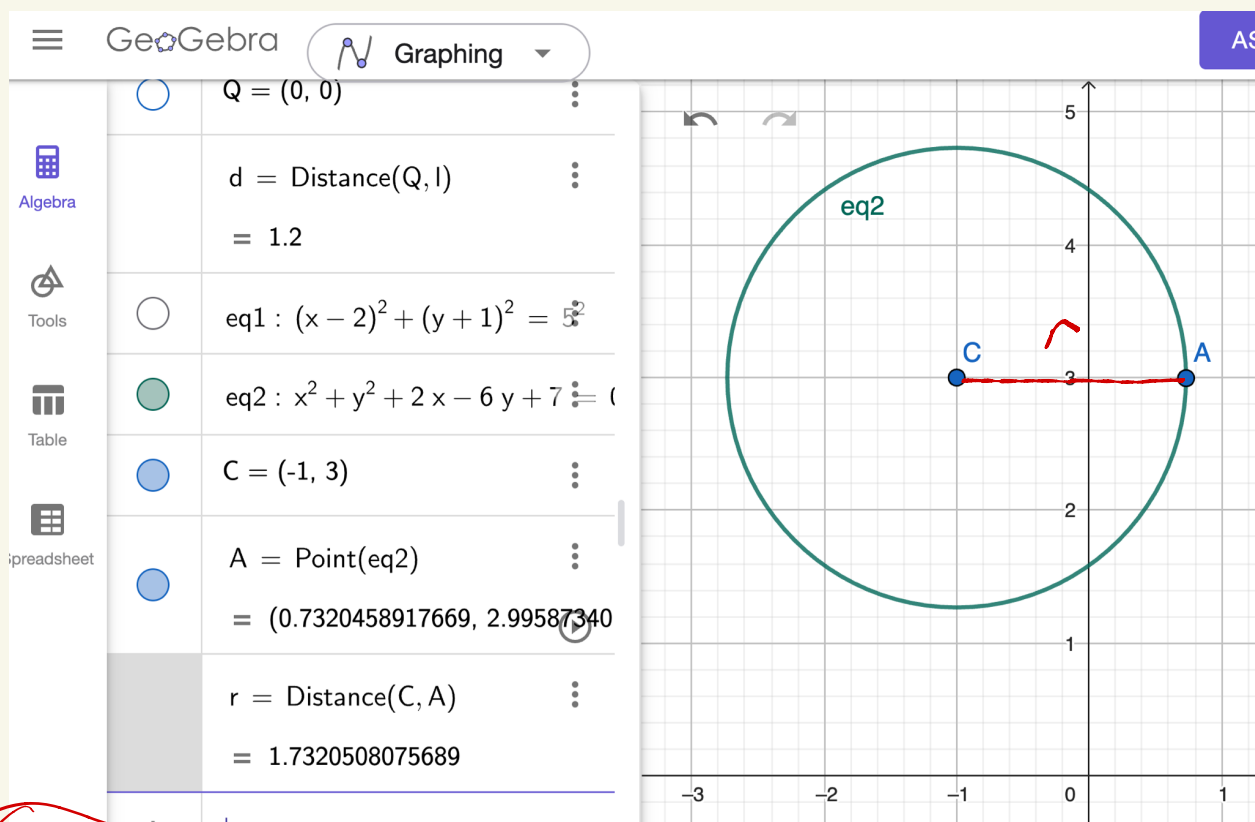
$$(x-3)^2 + (y-1)^2 = 53 \quad \text{Standard form}$$

$$x^2 - 6x + 9 + y^2 - 2y + 1 = 53 \quad \text{Expand the squares}$$

$$x^2 - 6x + y^2 - 2y = 43 \quad \text{Subtract 10 to get expanded form}$$

$$x^2 + ax + y^2 + by = c \quad \text{Expanded form}$$

↑ difficult.



check:

$$\begin{aligned}
 (x - (-1))^2 + (y - 3)^2 &= 3 \\
 (x + 1)^2 + (y - 3)^2 &= 3 \\
 x^2 + 2(1)x + (1)^2 + y^2 + 2(-3)y + (-3)^2 &= 3 \\
 x^2 + 2x + 1 + y^2 - 6y + 9 &= 3 \\
 x^2 + 2x + y^2 - 6y + 10 - 3 &= 0 \\
 x^2 + y^2 + 2x - 6y + 7 &= 0
 \end{aligned}$$

simple.

$$1 = \left(\frac{2}{2}\right)^2$$

$$(x+1)^2 = (x+1)(x+1) = x^2 + \underline{x} + \underline{x} + \underline{1} = x^2 + 2x + 1$$

$$(y-3)^2 = (y-3)(y-3) = y^2 - 3y - 3y + (-3)^2 = y^2 - 6y + 9$$

Simple

$$\begin{aligned}
 & \left(x + \frac{2}{2}\right)^2 \\
 &= \underline{\left(x + 1\right)^2} \\
 &= (x+1)(x+1) \\
 &= x^2 + x + x + 1^2 \\
 &= \underline{x^2 + 2x} + \underline{\left(\frac{2}{2}\right)^2}
 \end{aligned}$$

Simple

$$\begin{aligned}
 & \left(x + \frac{b}{2}\right)^2 \\
 &= (x+a)^2 \\
 &= (x+a)(x+a) \\
 &= x^2 + ax + ax + a^2 \\
 &= x^2 + 2ax + a^2 \\
 &= x^2 + \underline{bx} + \underline{\left(\frac{b}{2}\right)^2}
 \end{aligned}$$

$b = 2a$
 $a = \left(\frac{b}{2}\right)$
 Difficult.

$$\begin{aligned}
 & \left(y + \left(\frac{-6}{2}\right)\right)^2 \\
 &= (y + (-3))^2 \\
 &= (y + (-3))(y + (-3)) \\
 &= y^2 - 3y - 3y + (-3)^2 \\
 &= y^2 - 6y + \left(\frac{-6}{2}\right)^2
 \end{aligned}$$

EXAMPLE 13 Identifying an Equation of a Circle

Show that the equation $x^2 + y^2 + 2x - 6y + 7 = 0$ represents a circle, and find the center and radius of the circle.

SOLUTION

Difficult

$$\begin{aligned}
 x^2 + y^2 + 2x - 6y + 7 &= 0 \\
 x^2 + 2x + \left(\frac{2}{2}\right)^2 + y^2 - 6y + \left(\frac{-6}{2}\right)^2 &= -7 + \left(\frac{2}{2}\right)^2 + \left(\frac{-6}{2}\right)^2 \\
 x^2 + 2x + (1)^2 + y^2 - 6y + (-3)^2 &= -7 + 1 + 9 \\
 (x + 1)^2 + (y - 3)^2 &= 3 \quad h = -1, k = 3 \\
 (x - (-1))^2 + (y - 3)^2 &= 3 \quad r^2 = 3 \\
 (x - h)^2 + (y - k)^2 &= r^2 \quad r = \sqrt{3}
 \end{aligned}$$

simple

∴ The centre is $C(h, k) = C(-1, 3)$ and the radius is $r = \sqrt{3}$

≈ 1.73

PLANE CURVES AND PARAMETRIC EQUATIONS

If f and g are functions defined on an interval I , then the set of points $(f(t), g(t))$ is a plane curve. The equations

$$\underline{x = f(t)} \quad \underline{y = g(t)}$$

where $t \in I$, are parametric equations for the curve, with parameter t .

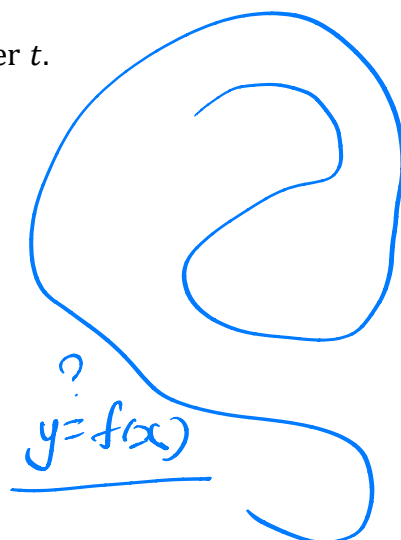
$$t \in \mathbb{R}$$

EXAMPLE 14 Sketching a Plane Curve

Sketch the curve defined by the parametric equations

$$\underline{x = t^2 - 3t} \quad \underline{y = t - 1}$$

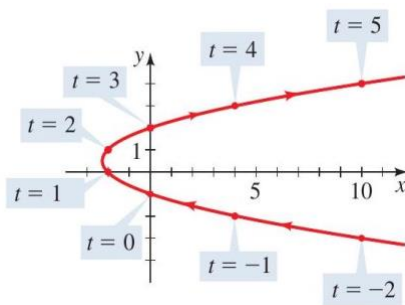
SOLUTION For every value of t we get a point on the curve. For example, if $t = 0$, then $x = 0$ and $y = -1$, so the corresponding point is $(0, -1)$. In Figure 1 we plot the points (x, y) determined by the values of t shown in the following table.



t	x	y
-2	10	-3
-1	4	-2
0	0	-1
1	-2	0
2	-2	1
3	0	2
4	4	3
5	10	4

$$\begin{cases} x = t^2 - 3t \\ y = t - 1 \end{cases}$$

parametric equations.



?
 $y = f(x)$
 $F(x, y) = 0$

EXAMPLE 15 Eliminating the Parameter

Eliminate the parameter in the parametric equations of Example 14.

SOLUTION

$$\begin{cases} x = t^2 - 3t \\ y = t - 1 \end{cases}$$

$$t = y + 1$$

$$x = (y+1)^2 - 3(y+1)$$

$$x = y^2 + 2y + 1 - 3y - 3$$

$$x - y^2 + y + 2 = 0$$

Cartesian equation:

